ON GRAFFITI/WOWII CONJECTURE 133: SMALL-PATH MASS BOUNDS AND A REDUCTION

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ABSTRACT. Conjecture 133 of *Written on the Wall II* states that every connected graph G on at least two vertices with no 4-cycle subgraph satisfies $p(G) \geq r(G) + \ell(G)$. We call this statement REAL-133. Here $p(G)$ is the largest order of an induced path, $r(G)$ is the radius, and $\ell(G) = S(G)/n$ is the average of $\alpha(G[N(v)])$.

For connected strong C_4-free P_6-free graphs we prove $S \leq 3n$, with equality only for the Petersen graph. In that class, $\ell > 5/2$ holds only for the Petersen graph and the graph obtained by deleting one vertex, with values 3 and 8/3. We prove REAL-133 when $\ell \leq 3$, when $r \leq 2$ and $\ell \leq 4$, when $r = 4$ and $\ell \leq 4$, and when $r \geq 5$ and $\ell \leq 4$. The case $r = 4$ and $p = 7$ gives the stronger bound $\ell \leq 2$. The only unsettled case with $\ell \leq 4$ is $r = 3$, $p = 6$ and $3 < \ell \leq 4$.

A theorem of Erdős, Saks and Sós reduces any counterexample to a radius-free bound GEN_K or a radius-extremal bound $\mathrm{RAD}(r)$. We prove the rooted bound of order 3, and the rooted bound of every order for connected strong C_4-free graphs whose blocks are bipartite or cycles. Explicit examples limit several strengthenings. The bounds GEN_K for $K \geq 3$, $\mathrm{RAD}(r)$ for $r \geq 5$, the rooted bound of order at least 4 outside that block class, unrestricted GEN_6, and REAL-133 remain open.

1. INTRODUCTION

All graphs are finite and simple. For a graph G with $n \geq 2$ vertices let $p(G)$ be the largest number of vertices of an induced path, $r(G)$ the radius, and

$$\ell(G) = \frac{1}{n} \sum_{v \in V(G)} \alpha\big(G[N(v)]\big),$$

where α is the independence number and $N(v)$ the open neighbourhood. Write $c_{C_4}(G) = 1$ if G contains no 4-cycle as a subgraph and $c_{C_4}(G) = 0$ otherwise.

Conjecture 1.1 (Graffiti/WOWII Conjecture 133)**.** *Every connected graph G on at least two vertices satisfies*

$$r(G) + \ell(G)^{\,c_{C_4}(G)} \leq p(G).$$

The conjecture is number 133 in E. DeLaVina's list *Written on the Wall II* of conjectures of the program Graffiti.pc [3]. We follow the formal statement in the Formal Conjectures repository [5], which reads `(rad : ℝ) + 1 G ^ cC4 ≤ (path G : ℝ)`. Earlier versions of that file had $\lfloor \ell(G) \rfloor$ in place of $\ell(G)$. Pull request #5727 of the repository [6] removed the floor. We work with the real-valued statement and call it REAL-133.

If G contains a 4-cycle, the conjecture reads $p(G) \geq r(G) + 1$. A shortest path between two vertices at distance $\mathrm{diam}(G) \geq r(G)$ is induced, so the inequality holds. Only graphs with no 4-cycle subgraph remain. We call them *strong C_4-free*. Since p and r are integers, REAL-133 for such a graph is equivalent to

$$p(G) - r(G) \;\geq\; \lceil \ell(G) \rceil. \tag{1}$$

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H. Chen [2] has studied mainly the floor version $p \geq r + \lfloor \ell \rfloor$. That preprint describes itself as not peer-reviewed, and we have not checked its proofs. For the real-valued form it records an exhaustive check on strong C_4-free graphs with at most 13 vertices, and its Theorem 6.7 states that $\operatorname{diam}(G) = 2$ and $\ell > 3$ imply $p \geq 6$. This overlaps the case $r = 2$, $3 < \ell \leq 4$ of our Theorem 4.4 for graphs of diameter 2. Its Lemma 7.18 contains the trace classification in the first part of our Lemma 3.4. Our proofs are independent of [2].

The arguments below are written in full. Section 3 gives the small-path mass bounds and their extremal graphs. Section 4 treats radius four, including the case $p = 7$. Section 5 is the reduction and a table of what remains. Section 7 is the rooted theorem for every index in the block class. Several inequalities studied by finite search are not theorems here: the repository does not contain a checker, an input list, and a completeness argument for them. They are listed in Section 11 and are not used as hypotheses.

2. Preliminaries

For $v \in V(G)$ let $d(v)$ be its degree and $t(v)$ the number of triangles containing v. Write $\lambda(v) = \lambda_G(v) = \alpha(G[N(v)])$, $S(G) = \sum_v \lambda(v)$, $e(G)$ for the number of edges and $T(G)$ for the number of triangles.

Lemma 2.1. *Let G be strong C_4-free. Then:*

1. *two distinct vertices have at most one common neighbour;*
2. *for every v, $G[N(v)]$ is a matching together with isolated vertices, so $\lambda(v)$ equals the number of components of $G[N(v)]$ and $\lambda(v) = d(v) - t(v)$;*
3. *$S(G) = 2e(G) - 3T(G)$.*

Proof. (1) Two common neighbours w, w' of x, y give the 4-cycle $xwyw'$. (2) A vertex of $N(v)$ with two neighbours in $N(v)$ would have two common neighbours with v. So the components are single vertices and edges, each edge being a triangle at v. (3) Sum $d(v) - t(v)$ over v. □

The components of $G[N(v)]$ are the *units* of v. For a vertex set U, $G-U$ is the subgraph induced by $V \setminus U$, and $D_G(U) = S(G) - S(G - U)$ is the *deletion loss* of U. We write $L(v) = D_G(\{v\})$.

Lemma 2.2. *Let G be strong C_4-free and $\emptyset \neq U \subseteq V(G)$. Then*

$$D_G(U) = \sum_{u \in U} \lambda(u) + \sum_{x \notin U} \#\{\text{units of } x \text{ contained in } U\}.$$

In particular $L(v) = 2\lambda(v) - t(v) = 2d(v) - 3t(v)$.

Proof. For $x \notin U$, deleting U removes exactly the units of x that lie inside U; a unit that is an edge with one end in U survives as a vertex. For $U = \{v\}$: the $2t(v)$ neighbours of v in triangles with v keep their value, and each of the other $d(v) - 2t(v)$ neighbours loses one. The total is $\lambda(v) + d(v) - 2t(v) = 2\lambda(v) - t(v)$. □

Lemma 2.3. *Let v be a cut vertex of G. Split the components of $G - v$ into two nonempty groups, and let G_1, G_2 be the subgraphs induced by the two groups together with v. Then $S(G) = S(G_1) + S(G_2)$.*

Proof. $G[N(v)]$ has no edges between the groups, so $\lambda_G(v) = \lambda_{G_1}(v) + \lambda_{G_2}(v)$. Every other vertex keeps its neighbourhood. □

We use three published results.

Theorem 2.4 (Erdős, Saks and Sós [4])**.** *Every connected graph G has an induced path on at least $2r(G) - 1$ vertices.*

Theorem 2.5 (Camby and Schaudt [1, Theorem 5])**.** *Let $k \geq 4$ and let G be a connected graph with no induced path on k vertices. Then G has a connected dominating set X such that $G[X]$ has no induced path on $k-2$ vertices or $G[X]$ is isomorphic to the cycle C_k.*

Theorem 2.6 (Seinsche [8])**.** *If a graph on at least two vertices has no induced path on four vertices, then it or its complement is disconnected.*

Theorem 2.5 is the implication (i)⇒(ii) of [1, Theorem 5] applied with $H = G$. We write P_k for the path on k vertices, and "G is P_k-free" means that G has no induced P_k, that is, $p(G) \leq k-1$. A connected graph of radius r has at least $2r$ vertices.

Lemma 2.7. *Let H be a connected strong C_4-free P_4-free graph on at least two vertices. Then H has a vertex adjacent to all other vertices.*

Proof. By Theorem 2.6, H is the join of two nonempty vertex sets A and B. If $|A|, |B| \geq 2$, then $a, a' \in A$ and $b, b' \in B$ span the 4-cycle $aba'b'$. So one side is a single vertex, and it is adjacent to everything. □

3. Small-path mass bounds

Lemma 3.1. *If G is connected and strong C_4-free with $r(G) = 1$, then $S(G) = 2(n-1) - m$, where m is the number of edges of $G[N(u)]$ for a dominating vertex u. Hence $\ell(G) < 2$, and (1) holds.*

Proof. Let $v \neq u$. By Lemma 2.1(1), v has at most one neighbour other than u, and such a neighbour lies in $N(u)$. So $G[N(v)]$ is a clique and $\lambda(v) = 1$; also $\lambda(u) = (n-1) - m$. Hence $S = 2(n-1) - m < 2n$ and $\lceil \ell \rceil \leq 2$. If G is complete then $G \in \{K_2, K_3\}$, $\ell = 1$ and $p = 2 = r + 1$. Otherwise two nonadjacent vertices of $N(u)$ together with u form an induced P_3, so $p \geq 3 \geq r + \lceil \ell \rceil$. □

Lemma 3.2. *If G is connected and strong C_4-free with $r(G) \geq 2$, then $p(G) \geq 4$.*

Proof. If G were P_4-free, Lemma 2.7 would give a dominating vertex, so $r(G) \leq 1$. □

Lemma 3.3. *If G is connected, strong C_4-free and P_5-free, then $S(G) \leq 2n - h$, where h is the number of vertices v with $\lambda(v) \geq 3$. In particular $S(G) \leq 2n$.*

Proof. Call v *simplicial* if $\lambda(v) = 1$ and *high* if $\lambda(v) \geq 3$. For $x \in N(v)$ put $X_x = N(x) \setminus N[v]$; then $N(x) \subseteq \{v\} \cup \{\bar{x}\} \cup X_x$, where $\bar{x}$ is the partner of x if x lies in an edge unit of v.

Key fact. Let x, y lie in different units of v. A common vertex of X_x and X_y would give the 4-cycle $vxwy$, so $X_x \cap X_y = \emptyset$. If $x' \in X_x$ and $y' \in X_y$ are nonadjacent, then $x'xvyy'$ is an induced P_5: $x \not\sim y$ because they lie in different units, $x \not\sim y'$ and $y \not\sim x'$ by disjointness, and v is adjacent to neither x' nor y'. So X_x is complete to X_y. If both are nonempty and $|X_x| \geq 2$, a vertex of X_y has two neighbours in X_x, which are also neighbours of x, a contradiction to Lemma 2.1(1). Hence if both are nonempty, then $|X_x| = |X_y| = 1$.

Main step. For each high v we find $\lambda(v) - 1$ simplicial vertices whose only high neighbour is v, one in each of $\lambda(v) - 1$ distinct units of v.

Case 1: some $u \in N(v)$ has $|X_u| \geq 2$. By the key fact $X_y = \emptyset$ for every y outside the unit of u. Then $N(y) \subseteq \{v, \bar{y}\}$, which is a clique, so y is simplicial; so is $\bar{y}$. Hence v is the only high neighbour of y, and the $\lambda(v) - 1$ units other than that of u supply the vertices.

Case 2: $|X_x| \leq 1$ for all $x \in N(v)$. Then $\lambda(x) \leq 2$ for $x \in N(v)$. Call a unit *active* if one of its vertices x has $X_x \neq \emptyset$. Suppose two units are active, with $y' \in X_y$ and $z' \in X_z$, and let x lie in a third unit. By the key fact $y' \sim z'$, and $xvyy'z'$ is an induced P_5: $x \not\sim y$ (different units), $x \not\sim y', z'$ (disjointness), $v \not\sim y', z'$, and $y \not\sim z'$ (disjointness). As $\lambda(v) \geq 3$, a third unit exists, so at most

one unit is active. Every vertex of an inactive unit is simplicial and its only high neighbour is v. This gives at least $\lambda(v) - 1$ units.

Counting. The chosen vertices for distinct high vertices are distinct, since each has a unique high neighbour. So $\sum_{v \text{ high}}(\lambda(v) - 1) \le \#\{\text{simplicial vertices}\}$, and

$$S = \sum_v \lambda(v) \le 2n - \#\{\text{simplicial}\} + \sum_{v \text{ high}} (\lambda(v) - 2) \le 2n - h. \qquad \square$$

Lemma 3.4. *Let G be connected, strong C_4-free and P_6-free with a dominating induced cycle $C = v_0v_1 \cdots v_5$, and let $W = V(G) \setminus V(C)$. Then every $w \in W$ is adjacent to exactly one antipodal pair $\{v_i, v_{i+3}\}$ and to no other vertex of C, distinct vertices of W use distinct pairs, and W is independent. Hence $|W| \le 3$, $S(G) = 12 + 4|W| < 3n$, and $r(G) \le 2$ if $W \ne \emptyset$.*

Proof. Indices are modulo 6. Let $w \in W$; it has a neighbour on C. If w has exactly one neighbour v_i, then $wv_iv_{i+1}v_{i+2}v_{i+3}v_{i+4}$ is an induced P_6. If $N(w) \cap C = \{v_i, v_{i+1}\}$, then $wv_{i+1}v_{i+2}v_{i+3}v_{i+4}v_{i+5}$ is an induced P_6. Two neighbours at distance 2 on C give a 4-cycle. A set of at least three vertices of C_6 contains two vertices at distance 2. So $N(w) \cap C$ is an antipodal pair. Two vertices of W with the same pair give two common neighbours. If $w \sim w'$ with distinct pairs, some $v_i \in N(w)$ and $v_j \in N(w')$ are adjacent on C, and wv_iv_jw' is a 4-cycle. There are no triangles, $e(G) = 6 + 2|W|$, and Lemma 2.1(3) gives $S = 12 + 4|W|$. Finally, if w uses $\{v_i, v_{i+3}\}$, then every vertex of C is within distance 2 of v_i (the vertex v_{i+3} through w), and each $w' \in W$ has a neighbour in $\{v_{i-1}, v_i, v_{i+1}\}$, because each of the three antipodal pairs meets this set. So v_i has eccentricity 2. $\square$

The same trace analysis applies to any vertex adjacent to an induced 6-cycle in a connected strong C_4-free P_6-free graph, whether or not the cycle dominates. We use that form in Proposition 3.7.

Lemma 3.5. *Let G be connected, strong C_4-free and P_6-free. Then either G has a vertex of eccentricity at most 2, or $G = C_6$.*

Proof. Apply Theorem 2.5 with $k = 6$ to get a connected dominating set X. If $G[X] \cong C_6$, Lemma 3.4 gives $G = C_6$ or a vertex of eccentricity 2. Otherwise $G[X]$ is P_4-free. If $|X| = 1$, the vertex in X dominates G. If $|X| \ge 2$, Lemma 2.7 gives $u \in X$ adjacent to all of $X \setminus \{u\}$, and since X dominates G, every vertex is within distance 2 of u. $\square$

Lemma 3.6. *If G is connected, strong C_4-free and P_6-free, then $S(G) \le 3n$, with equality if and only if G is the Petersen graph.*

Proof. If $G = C_6$ then $S = 12 < 18$. Otherwise, by Lemma 3.5, fix u of eccentricity at most 2. Put $N_1 = N(u)$, $N_2 = V \setminus N[u]$, and let m_1 be the number of edges of $G[N_1]$. By Lemma 2.1(1) each $c \in N_2$ has exactly one neighbour in N_1, its *parent*. For $a \in N_1$ let C_a be the set of its children. An edge cd of $G[N_2]$ with distinct parents is a *cross edge*. Let E' be the set of cross edges, and let D_c be the set of cross neighbours of c.

Local values. $\lambda(u) = |N_1| - m_1$. For $a \in N_1$ with partner $\bar{a}$ (if any), no child of a is adjacent to u or $\bar{a}$, so the units of a are $\{u, \bar{a}\}$ (or $\{u\}$) together with the components of $G[C_a]$; hence $\lambda(a) = 1 + \alpha(G[C_a])$. For $c \in C_a$: c has at most one neighbour $s \in C_a$ (two would be common neighbours of a and c), the vertices of D_c are not adjacent to a, and s is adjacent to no $d \in D_c$ (else c, a, s, d is a 4-cycle). So $\lambda(c) = 1 + \alpha(G[D_c])$. Using $\sum_a |C_a| = |N_2|$ and $n = 1 + |N_1| + |N_2|$ we get

$$3n - S = 3 + |N_1| + m_1 + \sum_a \big(|C_a| - \alpha(G[C_a])\big) + |N_2| - \sum_c \alpha(G[D_c]) \;\ge\; 3 + |N_1| + |N_2| - 2|E'|. \tag{2}$$

So it suffices to show $2|E'| \le 3 + |N_1| + |N_2|$.

Fact 1. Let $cd \in E'$ with $c \in C_y$ and $d \in C_z$. Then $y \not\sim z$, else $ycdz$ is a 4-cycle. Let $y' \in N_1 \setminus \{y, z\}$ and $x \in C_{y'}$. Then x is adjacent to c or d. Indeed, otherwise: if $y' \not\sim y$, then $xy'uycd$ is an induced P_6; if $y' \sim y$, then $y' \not\sim z$ (since $G[N_1]$ is a matching) and $xy'uzdc$ is an induced P_6. Two children of y' adjacent to the same vertex of $\{c, d\}$ would be two common neighbours of y' and that vertex. So $|C_{y'}| \leq 2$.

Fact 2. If $E' \neq \emptyset$, at most three vertices of N_1 have children. Suppose $cd \in E'$ has parents y, z, and $y_1, y_2 \in N_1 \setminus \{y, z\}$ have children x_1, x_2. Let $H = G[\{c, d, x_1, x_2\}]$. These four vertices have distinct parents, so every edge of H is a cross edge, and by Fact 1 each of the other two vertices is adjacent to an endpoint of that edge. Hence H has no induced $K_2 \cup K_1$. So the complement of H has no induced P_3, that is, it is a disjoint union of cliques, and H is complete multipartite. A complete multipartite graph on four vertices with at least one edge and no 4-cycle subgraph is $K_{1,3}$. Let its centre be c^* and its leaves a, b, d^*, with parents a_0, b_0, d_0. Since $G[N_1]$ is a matching, two of these parents are nonadjacent, say $a_0 \not\sim b_0$. Then a_0, u, b_0, b, c^*, d^* is an induced P_6: the leaves are pairwise nonadjacent, and each child is adjacent only to its own parent in N_1. This is a contradiction.

Counting. A child has at most one cross neighbour in each other class $C_{y'}$ (two would be common neighbours with y'), so its number of cross neighbours is at most $k - 1$, where $k \leq 3$ is the number of vertices of N_1 with children. If $E' = \emptyset$ we are done. If $k = 2$, every child has at most one cross neighbour, and $2|E'| \leq |N_2|$. Let $k = 3$, with classes C_1, C_2, C_3. If a child in C_i has two cross neighbours, in C_j and C_l, then Fact 1 applied to these two cross edges gives $|C_j|, |C_l| \leq 2$. Distinct such children of C_i have distinct neighbours in C_j (else two common neighbours with y_i), so C_i contains at most two children with two cross neighbours, and there are at most six in total. Hence $2|E'| \leq |N_2| + 6 \leq |N_2| + 3 + |N_1|$, because $|N_1| \geq k = 3$.

Equality. The Petersen graph is P_6-free: if $v_0 \cdots v_5$ were an induced path, then v_0 would have a common neighbour with each of v_3, v_4, v_5 (diameter 2); these lie off the path (the only path neighbour of v_0 is v_1, which is not adjacent to them) and are distinct (girth 5), but v_0 has only two neighbours off the path. Also $\lambda \equiv 3$, so $S = 30 = 3n$. Conversely, suppose $S = 3n$. Then $G \neq C_6$; the dominating-C_6 case is excluded because there $S < 3n$ (Lemma 3.4); and $E' \neq \emptyset$, because otherwise (2) gives $3n - S \geq 3 + |N_1| + |N_2| > 0$. If $k = 2$ then $3n - S \geq 3 + |N_1| > 0$. So $k = 3$, and equality throughout forces: $|N_1| = 3$; $m_1 = 0$; no edges inside any C_a; every D_c is independent; and exactly six children have two cross neighbours, two in each class. By Fact 1 every class has at most two children, so every class has exactly two, and every child has exactly two cross neighbours. The cross edges then form a 2-regular graph on the six children with no triangle (each D_c is independent), that is, a 6-cycle. Two children of the same parent at distance 2 on this cycle would have two common neighbours, and they are not adjacent, so they are antipodal. This determines G up to isomorphism, and the Petersen graph has exactly this structure: in it, the six vertices at distance 2 from a vertex induce a 6-cycle. □

Proposition 3.7. *Let G be connected, strong C_4-free and P_6-free, of order at least two. If $\ell(G) > 5/2$, then G is the Petersen graph P or the graph $P - v$ obtained by deleting one vertex of P. These graphs have $\ell(P) = 3$ and $\ell(P - v) = 8/3$. Every graph in the class other than P satisfies $\ell \leq 8/3$, with equality precisely for $P - v$.*

Proof. Throughout, $S = 2e - 3T$ by Lemma 2.1. A chordless cycle of length at least seven contains an induced P_6. Thus the only possible holes have lengths five and six.

Case 1: an induced six-cycle. Label its vertices $c_0, \ldots, c_5$, indices modulo 6, and write A for the set of vertices at distance one from the cycle. The trace argument in the proof of Lemma 3.4 shows that each $a \in A$ meets the cycle in one antipodal pair. Distinct vertices of A use distinct pairs, and A is an independent set: an edge of A joins distinct pairs, and some contact of one end is adjacent on the cycle to some contact of the other, giving a 4-cycle. Hence $k = |A| \leq 3$.

Every vertex x at distance two from the cycle is adjacent to every vertex of A. Such an x has at least one neighbour in A. If $x \sim a$ and $x \not\sim b$ for $a, b \in A$, choose contacts so that $c_i \in N(a)$ and $c_{i+2} \in N(b)$. This is possible: the pairs are distinct, so some contact of b lies at cycle distance two from some contact of a, and we orient the cycle from that contact of a toward that contact of b. The path $x, a, c_i, c_{i+1}, c_{i+2}, b$ is then an induced P_6. Indeed x meets the cycle in nothing, a meets the cycle only in its antipodal pair (neither c_{i+1} nor c_{i+2}), b meets the cycle only in its antipodal pair (so $b \sim c_{i+2}$ and $b \not\sim c_i, c_{i+1}$), and $a \not\sim b$.

There is no vertex at distance three or more from the cycle. A shortest path z, x, a, c_i from such a vertex down to the cycle, extended by c_{i+1}, c_{i+2}, is an induced P_6: the vertex z has no neighbour in A or on the cycle, and x has no neighbour on the cycle.

Let h be the number of vertices at distance two. If $k = 0$, then A is empty, so the cycle has no neighbour and $G = C_6$. If $k = 1$, every distance-two vertex is adjacent to the unique vertex $a \in A$. The distance-two vertices induce a matching together with isolated vertices: a vertex of that layer with two neighbours in the layer would have two common neighbours with a. Writing m for the number of edges of the matching gives $n = 7 + h$, $e = 8 + h + m$ and $T = m$, so $S = 16 + 2h - m$. Then

$$5n - 2S = 5(7 + h) - 2(16 + 2h - m) = 3 + h + 2m > 0,$$

hence $\ell < 5/2$.

If $k \in \{2, 3\}$, then $h \leq 1$: two vertices of the second layer would both be adjacent to two vertices of A. There are no triangles. The second-layer vertex, if it exists, is adjacent to every vertex of A and to nothing on the cycle, A is independent, and each vertex of A contributes two edges to the cycle. Thus $n = 6 + k + h$ and $S = 2(6 + 2k + kh)$. The four possibilities are

(k, h)	$(2, 0)$	$(2, 1)$	$(3, 0)$	$(3, 1)$
S/n	$5/2$	$8/3$	$8/3$	3

The graph with $(k, h) = (3, 1)$ is the equality case of Lemma 3.6: the second-layer vertex has three pairwise nonadjacent neighbours, each with two children, and those six children induce a 6-cycle on which children of the same parent are antipodal. That graph is the Petersen graph P. Deleting the second-layer vertex leaves $(3, 0)$. Deleting one vertex of A leaves $(2, 1)$. The Petersen graph is vertex-transitive, so both deletions are isomorphic to $P - v$. Lemma 3.6 records that P is P_6-free and strong C_4-free, and both properties pass to induced subgraphs. From the table, $\ell(P) = 3$ and $\ell(P - v) = 8/3$. The graph $(2, 0)$ has $\ell = 5/2$.

Case 2: an induced five-cycle and no induced six-cycle. Write $C = c_0 \cdots c_4 c_0$. Every vertex at distance one from C has a singleton or consecutive-pair trace: any other pair of contacts lies at distance two on C and forms a 4-cycle. If a vertex x were at distance two, with a neighbour a in the first layer, the path x, a followed by four consecutive cycle vertices, beginning at one contact of a and omitting the second contact if a has one, would be an induced P_6. Thus C dominates G.

Write h for the number of vertices with singleton trace and j for the number with a consecutive-pair trace. At most one vertex carries each of the five pair traces, since two would have two common neighbours, so $j \leq 5$. The edges outside C are of the following kinds.

- Two singleton vertices with different supports are nonadjacent. Neighbouring supports give a 4-cycle. Supports at distance two give an induced 6-cycle along the longer arc of C. Edges within one singleton class form a matching, because two neighbours in that class would be two common neighbours of the supporting cycle vertex. Let m be the number of these edges.

- No two pair vertices are adjacent. From their two distinct support edges one can choose adjacent cycle vertices, one from each edge, and those two vertices together with the pair vertices form a 4-cycle.
- A singleton supported at c_i meets only a pair vertex supported on the opposite edge $\{c_{i+2}, c_{i+3}\}$. Every other pair edge contains a neighbour of c_i and yields a 4-cycle. That pair vertex meets at most one vertex of the singleton class, again by the common-neighbour bound. Let b be the number of singleton-to-pair edges, so $b \le \min(h, j)$.

Consequently $n = 5 + h + j$ and $e = 5 + h + 2j + m + b$. The triangles are the j pair triangles and the m matching triangles, so $T = j + m$ and

$$S = 10 + 2h + j - m + 2b.$$

Therefore

$$3S - 7n = -5 - h - 4j - 3m + 6b.$$

The inequality $6\min(h, j) \le h + 5j$ holds because $6h \le h + 5j$ when $h \le j$ and $6j \le h + 5j$ when $j \le h$. Hence $6b \le h + 5j$, and

$$3S - 7n \le -5 - 3m + j \le -5 + j \le 0,$$

where the last step is $j \le 5$. Thus $\ell \le 7/3 < 5/2$.

Case 3: no holes. The graph is chordal, and every nonempty induced subgraph has a simplicial vertex. The neighbourhood of a simplicial vertex is a clique. The degree is at most two: four vertices of a clique contain a 4-cycle. A simplicial vertex of a connected chordal graph on at least two vertices is not a cut vertex, and the graph left by deleting it is chordal and strong C_4-free. In the graph present at the deletion, the loss is $L(v) = 2d(v) - 3t(v)$. Degree one gives loss 2. Degree two, with the two neighbours adjacent, gives one triangle and loss 1. Deleting vertices until a single vertex remains, one has $S \le 2(n - 1)$. Thus $\ell \le 2(1 - 1/n) < 5/2$.

The three cases give the proposition. The only graphs with $\ell > 5/2$ are P and $P - v$, arising in Case 1. Every other graph satisfies $\ell \le 8/3$, and the value $8/3$ occurs only for $P - v$. □

Remark 3.8. Karthick and Maffray give a structural decomposition for graphs with no induced P_6 and no C_4, and they discuss Petersen blow-ups and clique cutsets [7]. Their class is larger than the strong C_4-free class of Proposition 3.7. A bound on S for the pieces of such a decomposition still requires a gluing argument. We cite that work only as a comparison of scope. Priority of the gap relative to the literature would need a separate check, and none is claimed here.

4. Radius four

Lemma 4.1. *Let G be connected, strong C_4-free and P_8-free. If an induced 8-cycle dominates G and $r(G) \ge 4$, then $G = C_8$.*

Proof. Write $C = c_0c_1 \cdots c_7c_0$, with indices modulo 8. For an outside vertex y the trace $A_y = N(y) \cap V(C)$ is nonempty.

The trace A_y is not contained in any three consecutive vertices of C. Suppose $A_y \subseteq \{c_i, c_{i+1}, c_{i+2}\}$. The ends c_i and c_{i+2} are not both in A_y, since y, c_i, c_{i+1}, c_{i+2} would be a 4-cycle. Thus A_y is a singleton or a consecutive pair. A singleton $\{c_i\}$ gives the induced path

$$y, c_i, c_{i+1}, c_{i+2}, c_{i+3}, c_{i+4}, c_{i+5}, c_{i+6}.$$

A consecutive pair $\{c_i, c_{i+1}\}$ gives the induced path

$$y, c_i, c_{i-1}, c_{i-2}, c_{i-3}, c_{i-4}, c_{i-5}, c_{i-6},$$

in which the omitted contact c_{i+1} does not lie on the path. Either path is an induced P_8.

Fix a cycle vertex, say c_0. The vertices at cycle distance greater than two from c_0 are the three consecutive vertices c_3, c_4, c_5. The trace of any outside vertex meets the complementary set

$\{c_7, c_0, c_1, c_2, c_6\}$, so every outside vertex is within distance three of c_0. The same holds with c_0 replaced by any vertex of C.

Suppose w lies outside C, and write $A = N(w) \cap V(C)$. No two vertices of A are at cycle distance two, for they would form a 4-cycle with w. A set of vertices of C in which every two distinct vertices are consecutive has size at most two, and every set of size at most two that is a singleton or a consecutive pair is contained in a triple of consecutive vertices. Both possibilities have been excluded. Therefore A contains two vertices at cycle distance three or four. Rotating, take them to be c_0 and c_d with $d \in \{3, 4\}$.

The antipode of c_0 is c_4. If $d = 4$, the path $c_0 w c_4$ has length two. If $d = 3$, then $c_3 \sim c_4$, so $c_0 w c_3 c_4$ has length three. Every other vertex of C is already at cycle distance at most three from c_0. Every outside vertex is within distance three of c_0. Thus the eccentricity of c_0 is at most three, contradicting $r(G) \geq 4$.

Hence C has no vertex outside it, and $G = C_8$. □

Remark 4.2. The proof of Lemma 4.1 does not use a machine enumeration. An independent count, with the labelled cycle fixed, finds 32 admissible nonempty traces and 790 configurations on those traces, including the bare cycle. By order $8, \ldots, 13$ the configuration counts are $1, 32, 262, 384, 107, 4$. There are 73 unrooted isomorphism classes, and only the bare cycle has radius at least four. That count is not a hypothesis of Theorem 4.3.

Theorem 4.3. *Let G be connected and strong C_4-free with $r(G) = 4$.*

1. *If $p(G) = 7$, then $S(G) \leq 2n$, so $\ell(G) \leq 2$.*
2. *If $p(G) \geq 8$ and $\ell(G) \leq 4$, then $p(G) \geq r(G) + \ell(G)$.*

In particular RAD(4) *holds: every such graph with $p(G) = 7$ satisfies $\ell(G) \leq 3$.*

Proof. (2) is immediate: $p(G) \geq 8 = r(G) + 4 \geq r(G) + \ell(G)$.

For (1), Theorem 2.4 is not required, but it is compatible: $p(G) \geq 7$. Apply Theorem 2.5 with $k = 8$. There is a connected dominating set X such that $G[X]$ is P_6-free or $G[X] \cong C_8$.

If $G[X] \cong C_8$, Lemma 4.1 gives $G = C_8$. Then $S = 2n$ and $\ell = 2$.

If $G[X]$ is P_6-free, apply Lemma 3.5 inside $G[X]$. A vertex of eccentricity at most two in $G[X]$ has eccentricity at most three in G, because X dominates G, contradicting $r(G) = 4$. Therefore $G[X] = C_6$. Let $C = v_0 \cdots v_5$ be this dominating induced 6-cycle, indices modulo 6, and let $W = V(G) \setminus V(C)$.

Traces. As in the proof of Lemma 3.4, strong C_4-freeness allows only three kinds of trace $N(w) \cap C$: a single vertex, a consecutive pair $\{v_i, v_{i+1}\}$, or an antipodal pair. If some w has trace $\{v_i, v_{i+3}\}$, then v_i is within distance two of every vertex of C, hence within distance three of every vertex of G, a contradiction. So there are no antipodal traces. Let C_i be the set of vertices with trace $\{v_i\}$.

Witnesses. Fix i. Since $r(G) = 4$, some vertex is at distance at least four from v_{i+3}. Every vertex of C is within distance three of v_{i+3}, and a vertex $w \in W$ is at distance at most $1 + \min\{d_C(v_{i+3}, x) : x \in N(w) \cap C\}$ from v_{i+3}, which is at most three unless the trace of w is $\{v_i\}$. So there is $w_i \in C_i$ at distance four from v_{i+3}. Every neighbour $z \in W$ of w_i lies in C_i: if z had a neighbour among $v_{i+2}, v_{i+3}, v_{i+4}$, then $d(w_i, v_{i+3}) \leq 3$; if z had the neighbour v_{i-1} or v_{i+1}, then $w_i, v_i, v_{i\pm1}, z$ would be a 4-cycle. So $N(w_i) \subseteq \{v_i\} \cup C_i$.

Edges in W. Let $x, y \in W$ be adjacent.

- $x \in C_0$, $y \in C_{\pm1}$: $x, v_0, v_{\pm1}, y$ is a 4-cycle.
- $x \in C_0$, $y \in C_2$: $w_5, v_5, v_0, x, y, v_2, v_3, w_3$ is an induced P_8.
- $x \in C_0$, $y \in C_3$: $w_1, v_1, v_0, x, y, v_3, v_4, w_4$ is an induced P_8. By symmetry, vertices of C_i and C_j with $i \neq j$ are never adjacent.

- x with trace $\{v_0, v_1\}$ and $y \in C_j$: for $j \in \{0, 1\}$ the vertices v_0, v_1, x, y span a 4-cycle; for $j = 5$ (respectively $j = 2$) the cycle x, v_0, v_5, y (respectively x, v_1, v_2, y) is a 4-cycle; for $j = 3$, $w_5, v_5, v_0, x, y, v_3, v_2, w_2$ is an induced P_8, and $j = 4$ is symmetric.
- x, y with consecutive-pair traces $\{v_0, v_1\}$ and $\{v_j, v_{j+1}\}$: $j = 0$ gives two common neighbours; $j = \pm 1$ and $j = \pm 2$ give 4-cycles as above; $j = 3$ gives the induced P_8 $w_2, v_2, v_1, x, y, v_4, v_5, w_5$.

In each displayed path the non-edges follow from the traces and from $N(w_i) \subseteq \{v_i\} \cup C_i$. So every edge inside W joins two vertices of the same class C_i, and these edges form a matching.

Mass. Let a be the number of singleton-trace vertices, let b be the number of consecutive-pair vertices, and let m be the number of edges inside W. Then

$$n = 6 + a + b, \qquad e = 6 + a + 2b + m, \qquad T = b + m,$$

so $S = 2e - 3T = 2n - b - m \leq 2n$, and $\ell \leq 2$. □

Theorem 4.4 (Theorem A)**.** *Let G be connected and strong C_4-free with $n \geq 2$. Then $p(G) \geq r(G) + \ell(G)$ whenever $\ell(G) \leq 3$, whenever $r(G) \leq 2$ and $\ell(G) \leq 4$, whenever $r(G) = 4$ and $\ell(G) \leq 4$, and whenever $r(G) \geq 5$ and $\ell(G) \leq 4$. If $r(G) = 4$ and $p(G) = 7$, then $\ell(G) \leq 2$.*

Proof. Write $r = r(G)$ and $c = \lceil \ell(G) \rceil$. The last sentence is Theorem 4.3(1). For the inequality we need $p \geq r + c$ in the listed ranges.

- $r = 1$: Lemma 3.1.
- $r = 2$, $c \leq 2$: Lemma 3.2.
- $r = 2$, $c = 3$: if $p \leq 4$, then Lemma 3.3 gives $\ell \leq 2$, a contradiction. So $p \geq 5$.
- $r = 2$, $c = 4$: if $p \leq 5$, then Lemma 3.6 gives $\ell \leq 3$, a contradiction. So $p \geq 6$.
- $r = 3$, $c \leq 2$: Theorem 2.4 gives $p \geq 5$.
- $r = 3$, $c = 3$: if $p \leq 5$, then by Lemma 3.5 and $r = 3$ we have $G = C_6$, with $\ell = 2$, a contradiction. So $p \geq 6$. Lemma 3.6 alone would give only $\ell \leq 3$ for $p \leq 5$, which does not rule out $c = 3$.
- $r \geq 4$, $c \leq 3$: Theorem 2.4 gives $p \geq 2r - 1 \geq r + 3$.
- $r = 4$ and $\ell \leq 4$: Theorem 2.4 gives $p \geq 7$. If $p = 7$, Theorem 4.3 gives $\ell \leq 2$, so $p \geq r + \ell$. If $p \geq 8$, Theorem 4.3(2) gives the same inequality.
- $r \geq 5$ and $\ell \leq 4$: Theorem 2.4 gives $p \geq 2r - 1 \geq r + 4 \geq r + c$.

□

Equality in REAL-133 occurs for C_5 ($r = 2$, $\ell = 2$, $p = 4$), the Petersen graph ($r = 2$, $\ell = 3$, $p = 5$) and C_6 ($r = 3$, $\ell = 2$, $p = 5$). We do not characterise all equality cases.

5. The reduction and the residual cases

For integers $K \geq 3$ and $r \geq 4$ consider the following statements.

GEN$_K$: every connected strong C_4-free graph G with $p(G) \leq 2K$ satisfies $S(G) \leq Kn$.
RAD(r): every connected strong C_4-free graph with $r(G) = r$ and $p(G) = 2r - 1$ satisfies $\ell(G) \leq r - 1$.

Theorem 4.3 is the case $r = 4$ of RAD(r), in the stronger form $\ell \leq 2$.

Theorem 5.1 (Theorem B)**.** *Let G be a connected strong C_4-free graph that violates Conjecture 1.1. Then $\ell(G) > 3$. Put $K = \lceil \ell(G) \rceil - 1 \geq 3$. Then either*

(a) *$r(G) \leq K$ and $p(G) \leq r(G) + K \leq 2K$, so G violates* GEN$_K$*; or*
(b) *$r(G) = K + 1 \geq 4$ and $p(G) = 2r(G) - 1$, and $\ell(G) > r(G) - 1$, so G violates* RAD$(K + 1)$.

Proof. Write $r = r(G)$, $p = p(G)$ and $c = \lceil \ell(G) \rceil = K+1$. Then $\ell > 3$ by Theorem 4.4. A violation means $p \le r + c - 1 = r + K$. Theorem 2.4 gives $2r - 1 \le p \le r + K$, so $r \le K + 1$. If $r \le K$, then $p \le r + K \le 2K$, and $S > Kn$ because $\ell > K$. If $r = K + 1$, then $2r - 1 \le p \le 2K + 1 = 2r - 1$, and $\ell > K = r - 1$. □

Since GEN_K ignores the radius, it is stronger than alternative (a). The statement GEN_K might fail for large K even if REAL-133 holds; see Remark 7.4.

Proposition 5.2. *For $k \ge 3$ let θ_k consist of two vertices x, y joined by k internally disjoint paths xa_ib_iy. Then θ_k is strong C_4-free, $n = 2k + 2$, $r = 3$, $p = 6$ and $S = 6k = 3n - 6$. Thus $p - r - \ell = 6/n \to 0$.*

Proof. Every cycle consists of two complete branches and so has length 6; in particular there is no 4-cycle. There are no triangles, so $S = 2e = 6k$. Every vertex has eccentricity 3: $d(x, y) = 3$ and $d(a_i, b_j) = 3$ for $i \ne j$. The path $a_jxa_ib_iyb_l$ (with j, i, l distinct) is induced. Conversely, let P be an induced path. If P avoids x and y, it lies inside one edge a_ib_i. If P contains x but not y, then x has at most two neighbours on P, each followed on P by at most its b-vertex, so $|P| \le 5$. If P contains x and y, then every a_j on P is adjacent on P to x and every b_j to y, at most one branch lies entirely on P, and so P consists of x, y, at most one full branch and at most one further neighbour of each of x and y. Hence $|P| \le 6$. □

Thus no argument with a uniform positive margin in ℓ can prove GEN_3, already among graphs of radius 3.

Range	Status	Reason
$\ell \le 3$	proved	Theorem 4.4
$r \le 2$ and $\ell \le 4$	proved	Theorem 4.4
$r = 4$ and $\ell \le 4$	proved	Theorem 4.4; $p = 7$ forces $\ell \le 2$
$r \ge 5$ and $\ell \le 4$	proved	Theorem 4.4, by Theorem 2.4
$r = 3$, $p = 6$, $3 < \ell \le 4$	open	the only unsettled case with $\ell \le 4$
$\ell > 4$	open	Theorem 5.1: a violation of GEN_K or of $\mathrm{RAD}(r)$

TABLE 1. Residual status of REAL-133 for connected strong C_4-free graphs.

The open cell in Table 1 is forced by the proved bounds. Let $3 < \ell \le 4$, and suppose G violates REAL-133. Theorem 5.1 gives $K = 3$, so either $r \le 3$ and $p \le 6$, or $r = 4$ and $p = 7$ with $\ell > 3$. The second alternative contradicts Theorem 4.3. If $r \le 2$, Theorem 4.4 applies. Thus $r = 3$ and $p \le 6$. Theorem 2.4 gives $p \ge 5$. If $p \le 5$, Lemma 3.6 gives $\ell \le 3$. Hence $p = 6$. If instead $p \ge 7$, then $p \ge 7 \ge r + \ell$. Theorem 2.4 also excludes $p < 7$ when $r = 4$. For $3 < \ell \le 4$, Lemma 3.6 excludes $p \le 5$. The single remaining cell with $\ell \le 4$ is therefore $r = 3$, $p = 6$ and $3 < \ell \le 4$. The inequality GEN_3 would close that cell. It is not proved in this paper; see Appendix A and Section 11.

6. The rooted bound of order three

For a vertex s of a connected graph F let $p_s(F)$ be the largest order of an induced path starting at s. Write $D = d_F(s)$ and $m = e(F[N(s)])$.

Definition 6.1. For an integer $q \ge 3$, the rooted bound $\mathrm{ROOT}(q)$ asserts that every connected strong C_4-free graph F and every vertex s with $p_s(F) \le q$ satisfy

$$S(F) \le (q - 1)(|F| - 1) - (q - 3)D - m.$$

The hypothesis $p_s(F) \le q$ belongs in the statement. Without it the displayed inequality is false: if $q = 3$ and $F = C_5$, then $S(F) = 10$ and $m = 0$, while $2(|F| - 1) = 8$.

Lemma 6.2. ROOT(3) *holds. Explicitly, if F is connected and strong C_4-free and s is a vertex at which no induced P_4 starts, then $S(F) \le 2(|F| - 1) - m$.*

Proof. A vertex at distance 3 from s would end a geodesic, hence an induced P_4, starting at s. So $V = \{s\} \cup N_1 \cup N_2$ with N_i the i-th distance layer. Each $y \in N_2$ has a unique parent $a \in N_1$. A neighbour $z \in N_2$ of y with a different parent would give the induced path $sayz$. So y has at most one neighbour in N_2, a sibling (two siblings would be common neighbours of a and y), and $N(y)$ is a clique; thus $\lambda(y) = 1$. For $a \in N_1$ we have $N(a) \subseteq \{s, \bar{a}\} \cup C_a$, where C_a is the set of children of a and $\bar{a}$ is the neighbour of a in N_1, if any. Since s and $\bar{a}$ lie in one unit, $\lambda(a) \le 1 + |C_a|$. Finally $\lambda(s) = |N_1| - m$. Summing, $S \le (|N_1| - m) + (|N_1| + |N_2|) + |N_2| = 2(|F| - 1) - m$. □

For $q \ge 4$ the bound ROOT(q) is not proved for arbitrary strong C_4-free graphs. Section 11 records the precise limitation. The next section proves every index inside a restricted class.

7. Rooted bounds for every index

Let $\mathcal{B}$ be the class of connected strong C_4-free graphs in which every block is bipartite or a cycle. Edges are blocks, and cycles may be odd. The class contains every connected strong C_4-free bipartite graph and every connected strong C_4-free cactus. The block restriction is necessary for a uniform rooted bound: $K_{2,3}$ rooted in its two-vertex part has $p_s = 3$ and $S = 12 > 2(n-1) - m = 8$.

A rooted graph (G, s) with $p_s(G) = q$ and $\mathrm{ecc}(s) = q - 1$ is called *terminal*. For a terminal instance put $R = n - q + 1 - D \ge 0$, the number of vertices outside $N[s]$ and outside a root-to-farthest-vertex geodesic. Set $c_2 = 0$, $c_3 = 2$, $c_4 = 3$ and $c_q = 2\lfloor (q-1)/2 \rfloor$ for $q \ge 5$.

Theorem 7.1. (1) *Every terminal instance in $\mathcal{B}$ satisfies $S(G) \le 2(D + q - 2) - m + c_q R$.*

(2) *Every $F \in \mathcal{B}$ with a vertex s such that $p_s(F) \le q$ satisfies*

$$S(F) \le c_q(n-1) - (c_q - 2)D - m.$$

In particular ROOT(q) *holds in $\mathcal{B}$ for every $q \ge 3$. If every block is bipartite or a triangle, one may take $c_q = 2\lfloor (q-1)/2 \rfloor$ for all $q \ge 3$.*

Proof. Two terminal facts. An endpoint x of a longest rooted induced path is not a cut vertex: a neighbour of x in another component of $G - x$ would extend the path. If z is at distance $q-1$ from s and $v_0 = s, \dots, v_{q-1} = z$ is a geodesic, any neighbour $y \ne v_{q-2}$ of z must meet an earlier path vertex (else the rooted path extends); meeting v_i with $i \le q-4$ shortens $d(s, z)$, and meeting v_{q-3} gives a C_4; so $y \sim v_{q-2}$, and two such y would give a C_4. Hence $L(z) = 2d(z) - 3t(z) \le 2$. Moreover, if $x \ne s, z$, $G - x$ is connected and z survives, then $q - 1 \le d_{G-x}(s, z) \le p_s(G - x) - 1 \le q - 1$, so the instance stays terminal with the same q.

Endpoint losses. Let x be the endpoint of a longest rooted path of order q. It lies in a single block. If that block is a triangle the loss $L(x)$ is 1, if an edge 2, if a cycle of length at least 5 it is 4. If it is bipartite, let ℓ be the order of the final path segment inside it. Block-tree separation shows that the neighbours of x in this block can only be blocked by earlier path vertices in the same block; every neighbour other than the predecessor has a blocker in the colour class of x, the vertex two positions earlier cannot block (a C_4), and each position blocks at most one neighbour. So $d(x) \le \lfloor (\ell - 1)/2 \rfloor \le \lfloor (q-1)/2 \rfloor$, there are no triangles at x, and $L(x) \le c_q$ for $q \ge 5$.

Bases. For $q \le 4$ a nontrivial bipartite block would yield, by the same endpoint count, a rooted induced path of order at least 5, so these instances are cacti. For a C_4-free cactus with c_3 triangle blocks and c_L longer cycle blocks, $S = 2(n-1) + 2c_L - c_3$ and $m \le c_3$. For $q = 3$ no longer cycle occurs, and (1) follows. For $q = 4$ every longer cycle is a C_5 through s. A chosen length-3 geodesic uses at most one edge of each such C_5: if it used two edges and then left the cycle, replacing that

two-edge arc by the other three-edge arc would give an induced rooted path of order 5. So each C_5 contributes two vertices to R, $R \geq 2c_L$, and $[2(D+2) - m + 3R] - S = R - 2c_L + c_3 - m \geq 0$. For $q = 2$ the root is universal and $S = 2D - m$.

Induction on n. Let E be the set of endpoints of rooted induced paths of order q and z a farthest vertex, so $z \in E$. If some $x \in E \setminus \{z\}$ exists, delete it: $x \notin N[s]$, x is not a cut vertex, its loss is at most c_q, the instance remains terminal with the same q, D and m, and R drops by one. If $E = \{z\}$, delete z: the new rooted maximum is $q-1$, the eccentricity $q-2$, R is unchanged and the loss is at most 2; induction gives $S \leq 2(D+q-3) - m + c_{q-1}R + 2 \leq 2(D+q-2) - m + c_qR$. Deletion preserves membership in $\mathcal{B}$. This proves (1); the bipartite-or-triangle variant is identical.

(2) Attach a new path with $q-1$ edges at s. The result is terminal of index q, lies in $\mathcal{B}$, has order $n+q-1$, mass $S + 2(q-1)$ and root degree $D+1$, and m is unchanged. Substituting into (1) gives (2). Since $c_q \leq q-1$ and $n - 1 - D \geq 0$,

$$\begin{aligned} c_q(n-1) - (c_q - 2)D - m &= c_q(n - 1 - D) + 2D - m \\ &\leq (q-1)(n-1-D) + 2D - m \\ &= (q-1)(n-1) - (q-3)D - m, \end{aligned}$$

so ROOT(q) follows. □

A matching upper bound of shape $S \leq (r-1)n - 2r(r-3)$, for graphs in $\mathcal{B}$ of radius $r \geq 5$ and path order $2r-1$, has been proposed from a structural description of radius-extremal graphs. That description is not proved in this paper, and the radius bound in $\mathcal{B}$ is not claimed.

Lemma 7.2. *Let G be strong C_4-free and triangle-free, and let $x_0x_1 \cdots x_k$ be an induced path that cannot be extended at x_k. If $d(x_k) \geq 2$, then $k \geq d(x_k) + 1$. If moreover G is bipartite, then $k \geq 2d(x_k)$.*

Proof. Let $y \neq x_{k-1}$ be a neighbour of x_k. It is not on the path, and, since the path cannot be extended, y has a neighbour x_i with $i \leq k-1$. Here $i \neq k-1$ (a triangle) and $i \neq k-2$ (the 4-cycle $x_{k-2}x_{k-1}x_ky$), so $i \leq k-3$. A vertex x_i is adjacent to at most one neighbour of x_k. Hence the $d(x_k) - 1 \geq 1$ such neighbours need distinct blockers among $x_0, \ldots, x_{k-3}$, and $k - 2 \geq d(x_k) - 1$. If G is bipartite, the blocker has the colour of x_k, so $i \equiv k \pmod 2$ and $i \leq k-4$; there are $\lfloor k/2 \rfloor - 1$ such positions, and $\lfloor k/2 \rfloor \geq d(x_k)$. □

Proposition 7.3. *Every connected regular bipartite strong C_4-free graph satisfies* GEN_K *for every K and satisfies REAL-133. Every Moore graph of diameter* 2 *and girth* 5 *satisfies REAL-133.*

Proof. Let G be d-regular. If $d = 1$ then $G = K_2$, with $r = 1$, $\ell = 1$, $p = 2$ and $S = n$, and both statements hold. Let $d \geq 2$. As G is triangle-free, $\lambda \equiv d$ and $\ell = d$. Lemma 7.2 applied to a longest induced path gives $p \geq 2d + 1$, and Theorem 2.4 gives $p \geq 2r - 1$; so $p \geq r + d$ both when $r \leq d + 1$ and when $r \geq d + 1$. If $p \leq 2K$, then $2d + 1 \leq 2K$, so $S = dn < Kn$. A Moore graph of degree $d \geq 2$ is triangle-free and strong C_4-free with $r = 2$ and $\ell = d$, and the first part of Lemma 7.2 gives $p \geq d + 2 = r + \ell$. □

Remark 7.4. A Moore graph of degree 57, if it exists, has $n = 57^2 + 1 = 3250$ and $S = 57n$. It satisfies REAL-133 by Proposition 7.3, which only needs $p \geq 59$. The statement GEN_{57} holds for it with equality in the mass bound, but GEN_{56} asks for $p \geq 113$ because $S > 56n$. We do not know whether such a graph would have $p \leq 112$. If it did, it would refute GEN_{56} while satisfying REAL-133. Its existence would therefore not by itself separate the two statements, but it shows that a proof of every GEN_K is a different assertion from REAL-133.

8. Negative results

The examples in this section are explicit graphs.

Example 8.1. Let $s \geq 3$, $t \geq 3$ and $m \geq 1$ be integers. Let $G_{s,t,m}$ be the cycle $C_{2s} = c_0 \cdots c_{2s-1}$ with a pendant path of t edges at every cycle vertex, together with m internally disjoint paths c_0xzyc_1. Then $n = 2s(t+1) + 3m$, $e = 2s(t+1) + 4m$ and $T = 0$, so $e - n - T = m \geq 1$. For $(s,t,m) = (3,3,1)$ one has $n = 27$, $e = 28$, $r = 6$ and $p = 11$. These graphs satisfy REAL-133, since $\ell = 2e/n < 8/3$. They also show that $e = n + T$ fails for radius-extremal strong C_4-free graphs of every radius $r \geq 6$: here $r(G_{s,t,m}) = s + t$ and $p(G_{s,t,m}) = 2(s+t) - 1$.

Two vertices have at most one common neighbour, so G is strong C_4-free and triangle-free. An added path of length four does not shorten distances among the older vertices, and c_0 remains within distance two of every vertex of such a path, so the eccentricity of c_0 is still $s + t$. A vertex w of an added path meets the tip of the arm at c_s only through c_0 or c_1, hence at distance at least $s + t$. Thus $r = s + t$. For the path order, let B be the block formed by C and the added paths. An induced path P meets B in a subpath Q and can leave B only at an end of Q that is a cycle vertex, through at most t arm vertices. The same case division as for a single ear shows $|P| \leq 2s + 2t - 1$, using $t \geq 3$ and $s \geq 3$, and the cycle minus one vertex together with the arms at its two ends attains this order.

The last sentence of Example 8.1 abbreviates a case division which we now record. If Q avoids both c_0 and c_1, then $|Q| \leq 2s-2$. If Q contains exactly one of them, then either Q lies on the cycle with $|Q| \leq 2s - 1$, or $|Q| \leq 2s + 2$ with at most one cycle end, or $|Q| \leq 7$ with no cycle end. If Q contains both, the same three shapes occur with $|Q| \leq 2s-1$, $|Q| \leq 2s+2$ and $|Q| \leq 6$. Adding at most t arm vertices at each cycle end of Q gives $|P| \leq \max\{2s-1+2t,\ 2s+2+t,\ 7\} = 2s+2t-1$.

Example 8.2. Forbidding induced paths on eleven vertices and requiring $L(v) \geq 6$ at every vertex does not forbid an induced star $K_{1,6}$. Two Petersen graphs sharing a vertex give a triangle-free strong C_4-free graph on $n = 19$ vertices with $S = 60$, minimum deletion loss 6, $p = 9$ and $\lambda = 6$ at the shared vertex. For $b \geq 2$, gluing b copies of the Petersen graph along a common edge gives a 2-connected graph B_b with $n = 8b + 2$, $S = 28b + 2$, minimum loss 6, $p = 9$ and $\lambda = 2b + 1$ at the two hubs. Thus singleton losses do not bound λ. These graphs are not counterexamples to GEN_5: the eight non-hub vertices of one Petersen block have deletion loss $28 < 5 \cdot 8 + 1$.

For an oriented edge (u, v) of a connected strong C_4-free graph G, let q be the maximum order of an induced path whose first two vertices are u, v in that order, and let E be the set of terminal vertices of the induced paths of that order. The endpoint inequality $\mathrm{EP}(q)$ asserts $D_G(E) \leq q|E|$ for every such oriented edge of maximum order q. For a root s with $p_s(G) = q$, if E is the set of terminal vertices of the induced paths of order q starting at s, the rooted inequality $\mathrm{EP}^+(q)$ asserts $D_G(E) \leq (q-1)|E|$.

Proposition 8.3. *For every integer $q \geq 4$, $\mathrm{EP}(q)$ holds for all connected strong C_4-free graphs if and only if $\mathrm{EP}^+(q-1)$ does.*

Proof. Both graphs below remain strong C_4-free, so $S = 2e - 3T$ and the deletion loss of a vertex set is determined by the edges and triangles that meet it.

First assume $\mathrm{EP}^+(q-1)$, and let (u, v) be an oriented edge of maximum induced-path order q, with terminal set E. Delete $W = N[u] \setminus \{v\}$, and let H be the component of v in $G - W$. An induced path of G that starts with uv has no further vertex in $N[u]$, because the path is induced, so the remainder is an induced path of H starting at v. Conversely, every induced path of H starting at v extends by u to an induced path of G, since no vertex of H other than v is adjacent to u. Thus $p_v(H) = q - 1$, and the terminal set in H is still E.

Each $x \in E$ has at most one neighbour in W: any such neighbour is adjacent to u, and two of them would be two common neighbours of u and x. Every neighbour of x outside H lies in W. Let m_W be the number of edges between E and W, so $m_W \leq |E|$, and let T_{extra} be the number of triangles that meet E and are not contained in H. Every such triangle uses one of those edges.

Comparing edges and triangles,

$$D_G(E) = D_H(E) + 2m_W - 3T_{\text{extra}} \le D_H(E) + 2|E|.$$

The rooted inequality in H is $D_H(E) \le (q-2)|E|$, because $\mathrm{EP}^+(q-1)$ contributes the factor $(q-1)-1$. Therefore $D_G(E) \le q|E|$.

Conversely, assume $\mathrm{EP}(q)$, and let (H, v) be a connected strong C_4-free graph with $p_v(H) = q-1$ and terminal set E. Form G by adding a vertex u adjacent to v and, for each $x \in E$, a new vertex a_x adjacent exactly to u and to x. Every terminal x fails to be adjacent to v: an induced path of order $q-1 \ge 3$ from v to x has no chord vx. The vertex a_x has degree two. Any cycle through a_x uses both edges ua_x and a_xx, and then continues from x to the other neighbour of u on the cycle. That other neighbour is v or some a_y. In the first subcase the x–v path in H has length at least two, so the cycle has length at least five. In the second the cycle has the form $u, a_x, x, \dots, y, a_y$ and again has length at least five. A cycle through u that avoids every a_x would need two neighbours of u in H, but the only neighbour of u in H is v. Thus G is strong C_4-free. No new triangle appears: the two neighbours of a_x are nonadjacent, and the neighbours of u are pairwise nonadjacent.

An induced path that starts with uv contains no a_x. Otherwise the edge ua_x would be a chord, since a_x is not the successor of u on that path. The induced uv-paths are therefore the paths u followed by an induced path of H starting at v. Their maximum order is q, and their terminal set is E. Exactly $|E|$ new edges meet E, namely the edges a_xx, and no new triangle meets E. Hence $D_G(E) = D_H(E) + 2|E|$. The oriented inequality gives $D_G(E) \le q|E|$, so $D_H(E) \le (q-2)|E|$, which is $\mathrm{EP}^+(q-1)$. □

Example 8.4. EP(9) fails. Let G have vertices $0, \dots, 14$ and adjacency lists

0: 4, 8, 11	1: 5, 8, 10	2: 6, 8, 9	3: 7, 10
4: 0, 9, 10	5: 1, 9, 11	6: 2, 10, 11	7: 3, 11
8: 0, 1, 2, 12	9: 2, 4, 5	10: 1, 3, 4, 6, 13	11: 0, 5, 6, 7, 14
12: 8, 13, 14	13: 10, 12	14: 11, 12.	

The graph is triangle-free and strong C_4-free, with $n = 15$, $e = 23$ and $S = 46$. For the oriented edge $(12, 8)$ the maximum order is $q = 9$, attained only by

$$12, 8, 2, 9, 4, 10, 3, 7, 11 \qquad \text{and} \qquad 12, 8, 2, 9, 5, 11, 7, 3, 10.$$

Thus $E = \{10, 11\}$ and $D_G(E) = 20 > 18$. The same computation gives a failure of $\mathrm{EP}^+(8)$ on the subgraph induced by $0, \dots, 11$, rooted at 8: $p_8 = 8$, $E = \{10, 11\}$ and $D(E) = 16 > 14$. The inequality EP(8) is not settled in this paper, nor is $\mathrm{EP}^+(6)$.

Example 8.5. Let $k = p - r$. Call an induced subgraph H of G an *eligible core* if it is proper, connected and dominating, and $p(H) \le p(G) - 2$. The budget CB asks for an eligible core with

$$S(G) - S(H) \le k\big(n - |H|\big) + |H|.$$

Together with REAL-133 for H and an induction on the order, the budget would imply REAL-133. The Petersen graph has $S = 30$, $p = 5$, $r = 2$ and $k = 3$, hence $\ell = 3$. An eligible core has $p(H) \le 3$. A connected triangle-free strong C_4-free P_4-free graph is a star, by Lemma 2.7 and the absence of triangles. In the Petersen graph a star with $k \le 3$ leaves has centre of degree 3. The vertices it dominates are the centre, its three neighbours, and at most two further neighbours of each leaf, at most $4 + 2k$ vertices in all. This is 10 only for $k = 3$, so the eligible cores are exactly the ten closed neighbourhoods $N[v]$. Each has $S(H) = 6$, and $S(G) - S(H) = 24 > 3 \cdot 6 + 4 = 22$.

The unrestricted budget therefore fails. The example has $\ell = 3$. It does not determine a cutoff such as $\ell > 5$, and this paper contains no census of eligible cores.

Example 8.6. The degree, triangle and distance data of the Heawood graph ($n = 14$, cubic, $T = 0$, $r = 3$, $S = 42$, and $Z = 28$ nonadjacent pairs with no common neighbour) satisfy the counting identities used above, including $\sum_v d(v)^2 = n(n-1) + 6T - 2Z$, together with the lower bounds $p \geq 2r - 1$ and $p \geq \min_v \lambda(v) + 2$, if p is replaced by the value 5. But $5 < r + \ell = 6$. This system of scalar constraints does not imply REAL-133. The Heawood graph itself has $p = 7$.

Example 8.7. Let F be the incidence graph of the projective plane $\mathrm{PG}(2, q)$, with $f = |F| = 2(q^2 + q + 1)$ and $S(F) = (q + 1)f$, and choose a root of F. Let $t = f - 1$ and $b \geq 1$. Take a 6-cycle $c_0 \cdots c_5$, attach a pendant path with t edges at each cycle vertex, and attach b copies of F at c_0 by identifying their roots with c_0. The result is strong C_4-free, because each new block meets the rest in a cut vertex, and

$$n = 6f + b(f - 1), \qquad S = 12f + b(q + 1)f, \qquad r = t + 3, \qquad p = 2t + 5 = 2r - 1.$$

Distances inside the flower are unchanged. Every vertex of an attached copy is within distance $f - 1 \leq t$ of c_0, so each cycle vertex has eccentricity $t + 3$, while a vertex x of a copy has distance $d_F(x, c_0) + t + 3$ from the tip of the arm at c_3. An induced path meets the cycle in at most five consecutive vertices, and each of its two ends outside the cycle contributes at most t vertices. Mass is additive at cut vertices by Lemma 2.3. For fixed q, as $b \to \infty$ one has $\ell \to (q + 1)f/(f - 1)$. Letting $q \to \infty$, with $r = f + 2$, this limit grows like $\sqrt{r/2}$. For $q = 5$ and $b = 17$ one has $n = 1409$, $r = 64$, $p = 127$ and $S = 7068 > 5n$. These graphs satisfy $\mathrm{RAD}(r)$ for their own radius, and they show that ℓ is unbounded on the class $p = 2r - 1$. Any theorem $\mathrm{RAD}(r)$ must use a bound that depends on r.

9. Relation to earlier work

Three published theorems are quoted in Section 2 and used as stated. Erdős, Saks and Sós proved that every connected graph has an induced path on at least $2r(G) - 1$ vertices [4]. With the bound $n \geq 2r$, this is the reduction of Theorem 5.1. Camby and Schaudt characterised the absence of a long induced path by a connected dominating set whose induced subgraph is a shorter path-free graph or a cycle [1]. The radius-four argument applies that theorem and then classifies traces by hand. Seinsche's theorem on graphs with no induced P_4 [8] is used for the small-mass bounds.

The conjecture is item 133 of DeLaVina's list [3]. We follow the real-valued reading fixed in [5, 6]. The comparison with Chen's preprint [2] is in the introduction. Remark 3.8 compares the extremal gap of Proposition 3.7 with the structure theory of Karthick and Maffray [7] and claims no priority.

10. Open problems

Theorem 4.4 and Theorem 5.1 leave the following questions open. Unrestricted GEN_6 and REAL-133 are among them.

(1) **The cell $r = 3$, $p = 6$, $3 < \ell \leq 4$.** This is the only unsettled case of REAL-133 with $\ell \leq 4$. It would follow from GEN_3. Appendix A reduces a minimum counterexample to three local statements that are not proved here.

(2) **The range $\ell > 4$.** A counterexample falls under GEN_K or $\mathrm{RAD}(K + 1)$ for $K = \lceil \ell \rceil - 1 \geq 4$. Neither family is proved for those indices. Inside $5 < \ell \leq 6$ the alternatives of Theorem 5.1 specialise to the pairs $(r, p) = (4, 9)$ and $(5, 10)$, together with the cells already closed by Theorem 4.4. Those two pairs are not settled, for bipartite graphs or in general. Section 11 explains why earlier partial arguments are not proofs.

(3) **Rooted bounds past order three, and $\mathrm{RAD}(r)$ for $r \geq 5$.** $\mathrm{ROOT}(3)$ is Lemma 6.2. In the class $\mathcal{B}$, Theorem 7.1 gives every index. For arbitrary strong C_4-free graphs, $\mathrm{ROOT}(q)$

is open for every $q \geq 4$, and RAD(r) is open for every $r \geq 5$. The case RAD(4) is Theorem 4.3.

(4) **Level 6.** Let $\mathcal{F}_0$ be the class of connected triangle-free strong C_4-free graphs G with $p(G) \in \{11, 12\}$, minimum degree at least 4, and with every endpoint of every induced path of order $p(G)$ of degree at most 7. Appendix B gives three proved implications about deletion. They do not show that every graph in $\mathcal{F}_0$ with $S > 6n$ has a nonempty set U of deletion loss at most $6|U|$, and they do not prove $S \leq 6n$ for every triangle-free strong C_4-free graph.

Conjecture 10.1. ROOT(q) *holds for every $q \geq 3$. For every $r \geq 5$, every connected strong C_4-free graph of radius r and path order $2r-1$ satisfies $S \leq (r-1)n - 2r(r-3)$, with equality only for C_{2r}.*

The case $q = 3$ is Lemma 6.2, and every q holds in $\mathcal{B}$ by Theorem 7.1. The radius bound is not claimed for $r \geq 5$.

Conjecture 10.2. *If F is connected and strong C_4-free, $p_s(F) = q$ and the eccentricity of s is $q-1$, then*

$$S(F) \leq (q-1)(n-1) - (q-3)D - m - (q-2)(q-3).$$

We do not derive ROOT(q) from Conjecture 10.2 in this paper. The terminal bound in $\mathcal{B}$ is the first part of Theorem 7.1.

Conjecture 10.3. *Every connected strong C_4-free graph G that is not a cycle and has $\ell(G) > 5$ has an eligible core H, in the sense of Example 8.5, with*

$$S(G) - S(H) \leq (p-r)\big(n - |H|\big) + |H|.$$

Example 8.5 shows that the budget without a restriction on ℓ is false, and that the Petersen graph is a counterexample with $\ell = 3$. The example does not show that 5 is the correct cutoff. If the conjecture, the range $\ell \leq 5$, and an induction on the order were all available, REAL-133 would follow. The range $\ell \leq 5$ is not a theorem, so this route is not a proof of REAL-133. Example 8.6 shows that a proof cannot rely only on the scalar constraints listed there.

11. Bounds not proved in this paper

A finite search is not a proof in this article unless the search is replaced by a written argument. This repository does not contain a versioned archive of a checker, its inputs, and a completeness argument for the inequalities below. Stored verdict files, and hashes of those files, authenticate recorded marks. They do not establish the inequalities. None of the following is a hypothesis of a theorem in this paper.

(1) GEN_3 and GEN_4, and GEN_K for every $K \geq 5$, are not proved. Appendix A is a conditional reduction for GEN_3 only.
(2) RAD(r) is not proved for any $r \geq 5$. The case $r = 4$ is Theorem 4.3.
(3) ROOT(q) is not proved for any $q \geq 4$ outside the class $\mathcal{B}$. Any correct statement of ROOT(q) includes the hypothesis $p_s(F) \leq q$, as in Definition 6.1.
(4) The bipartite form of GEN_5 is not proved. In particular the pairs $(r, p) = (4, 9)$ and $(5, 10)$ are not closed for bipartite graphs, and they are not closed in general. No argument in this paper uses an absent classification of 2-connected P_6-free blocks, an absent list of traces, or the unproved bipartite bound to declare either pair closed.
(5) No structure theorem for connected strong C_4-free graphs of radius $r \geq 5$ and path order $2r-1$ is proved here. The radius bound in the class $\mathcal{B}$ that would depend on such a theorem is not claimed.

(6) Example 8.5 does not count eligible cores. A reported count of cores of order at most 12 is not a result of this paper and is not used to justify the cutoff in Conjecture 10.3.
(7) Unrestricted GEN_6, with or without triangles, is open. REAL-133 is open.

Appendix A. A conditional reduction for GEN_3

The three local statements named at the end of this appendix are not proved. With them, the hand arguments below would give GEN_3. Without them, GEN_3 remains open, and so does the cell $r = 3$, $p = 6$, $3 < \ell \le 4$.

Theorem A.1. *Let G be a counterexample to GEN_3 of minimum order: G is connected and strong C_4-free, $p(G) \le 6$, $S(G) \ge 3n+1$, and n is minimum. Then G is 2-connected; $S(G)-S(G-U) \ge 3|U|+1$ for every nonempty proper $U \subset V(G)$; $2\lambda(v) - t(v) \ge 4$ for every v, so $\lambda(v) \ge 2$; and every vertex with $\lambda(v) = 2$ has degree 2 and nonadjacent neighbours, and no two vertices of degree 2 are adjacent.*

Proof. Since $S \le 2 < 3n$ for $n \le 2$, one has $n \ge 3$. Let v be a cut vertex. For each component C of $G-v$ let ρ_C be the largest order of an induced path in $G[C \cup \{v\}]$ with end v. Two such paths from different components form an induced path on $\rho_C + \rho_{C'} - 1 \le 6$ vertices, so some component C_1 has $\rho_{C_1} \le 3$. Put $G_1 = G[C_1 \cup \{v\}]$ and $G_2 = G - C_1$, with $n_i = |G_i|$, so $n = n_1 + n_2 - 1$ and $n_1 \ge 2$. Both graphs are connected, smaller than G, and P_7-free. Lemma 6.2 gives $S(G_1) \le 2(n_1 - 1)$, and minimality gives $S(G_2) \le 3n_2$. By Lemma 2.3, $S(G) \le 2(n_1 - 1) + 3n_2 = 3n - (n_1 - 1) < 3n$.

Every component H of $G - U$, for U nonempty and proper, is connected, smaller, and P_7-free, so $S(H) \le 3|H|$. Summing, $S(G - U) \le 3(n - |U|)$, and $S(G) \ge 3n + 1$ gives the deletion bound. The case $U = \{v\}$ and Lemma 2.2 give $2\lambda(v) - t(v) \ge 4$. If $\lambda(v) = 2$, then $t(v) = 0$, so $d(v) = 2$ and the neighbours are nonadjacent. Every degree-2 vertex has $\lambda = 2$ and hence lies in no triangle. If $u \sim w$ both have degree 2, with other neighbours x and y, then $x \ne y$ because $t(u) = 0$. The set $\{u\}$ is a unit of x and $\{w\}$ is a unit of y, so Lemma 2.2 with $U = \{u, w\}$ gives loss $2 + 2 + 1 + 1 = 6 < 7$. □

The same mass bound for a disconnected graph is the sum of the bounds on its components. An isolated vertex has $S = 0$. Call a graph *admissible* if it is strong C_4-free and P_7-free, $L(v) \ge 4$ for every vertex, and no two adjacent vertices have degree 2. A minimum counterexample is admissible. In an admissible graph, $\lambda(v) = 2$ forces $t(v) = 0$ and $d(v) = 2$. Write B for the vertices of degree at least 3, the *branch vertices*, and Z for the degree-2 vertices, the *ears*. Put $D = G[B]$, $d_D(v) = |N(v) \cap B|$ and $q(v) = |N(v) \cap Z|$. All triangles lie in D.

Lemma A.2. *In an admissible graph, $q(v) \le 4$ for every vertex v.*

Proof. Let u_i be ears at v with other neighbours a_i. The a_i are distinct, nonadjacent to v, and of degree at least 3. Put $E_i = N(a_i) \setminus \{u_i\}$. Then $|E_i| \ge 2$ and $E_i \cap N[v] = \emptyset$.

If $a_i \not\sim a_j$, then $E_i = \{w, p\}$ and $E_j = \{w, q\}$ with $pq \in E$, $w \not\sim p$ and $w \not\sim q$. Indeed two common neighbours of a_i and a_j are forbidden, so $|E_i \cap E_j| \le 1$. Since $|E_i| \ge 2$, the private part $E_i \setminus E_j$ is nonempty, and likewise $E_j \setminus E_i$. If p lies in the first private part and q in the second with $p \not\sim q$, the path $pa_iu_ivu_ja_jq$ is induced. Thus the private parts are complete to each other. If one of them contained two vertices, those two vertices and one vertex of the other private part would form a 4-cycle with the corresponding a-vertex. Both private parts therefore have size one, and $|E_i \cap E_j| = 1$. An edge from the common vertex to a private vertex is another 4-cycle. In particular a_i and a_j have degree 3 and lie in no triangle.

No three of the vertices a_i are pairwise nonadjacent. Otherwise the sets E_i meet pairwise in a single vertex. If the three intersections are the three pairs of a triple, the claim forces a triangle on that triple, and one a_i together with two vertices of the triple forms a 4-cycle. Otherwise

$E_i = \{w, p_i\}$ with $p_1p_2p_3$ a triangle and w nonadjacent to each p_i, and $vu_1a_1wa_2p_2p_3$ is an induced P_7.

Five ears are impossible. The graph F they determine on $\{a_1, \dots, a_5\}$ has independence number at most two. If some vertex a of F is adjacent to the other four, those four induce a matching. The independence number forces that matching to be perfect, say $\{b_1b_2, c_1c_2\}$. Then $b_1 \not\sim c_1$, so the preceding paragraph says that b_1 lies in no triangle, but ab_1b_2 is a triangle. Otherwise every vertex of F has a non-neighbour, hence has degree 3 in G and lies in no triangle. One neighbour of each such vertex is its ear, so F has maximum degree at most two. It is triangle-free and has no 4-cycle, and its independence number is at most two, so F is a 5-cycle. An ear at v together with four vertices of that cycle is then an induced P_7. □

Proposition A.3. *Let G be admissible, with B, Z, D, d_D and q as above. Suppose that $d_D(v) \le 3$ for every v, that $d_D(v) = 3$ implies $q(v) = 0$, and that $d_D(v) = 2$ with the two neighbours in B nonadjacent implies $q(v) \le 2$. Then $S(G) \le 3n$.*

Proof. Ears have nonadjacent neighbours and Z is an independent set, so $S(G) = S(D) + 4|Z|$. Since $|Z| = \frac{1}{2}\sum_{v\in B} q(v)$ and $n = |B| + |Z|$,

$$S - 3n = \sum_{v\in B}\Big(d_D(v) - t_D(v) - 3 + \tfrac{1}{2}q(v)\Big).$$

Lemma A.2 gives $q(v) \le 4$. If $d_D(v) = 2$ and $t_D(v) = 0$, the two neighbours in B are nonadjacent, for an edge between them would be a triangle. The three suppositions and the ear bound make every summand nonpositive: the summand is at most -1 when $(d_D, t_D) = (0, 0)$, at most 0 when $(d_D, t_D) = (1, 0)$, at most 0 when $(d_D, t_D) = (2, 0)$, at most 0 when $(d_D, t_D) = (2, 1)$, and equal to $-t_D(v) \le 0$ when $d_D(v) = 3$. The suppositions exclude $d_D(v) \ge 4$. □

The three suppositions in Proposition A.3 are not proved in this paper. Theorem A.1 shows that a minimum counterexample to GEN_3 is admissible, so those suppositions would imply GEN_3. They do not, by themselves, make GEN_3 a theorem.

Appendix B. Deletion in the triangle-free class

Let $\mathcal{F}$ be the class of finite triangle-free strong C_4-free graphs with $p \le 12$. Write $\varepsilon_6(G) = S(G) - 6|G|$, and call G *positive* when $\varepsilon_6(G) > 0$. A nonempty set $U \subseteq V(G)$ is 6-*cheap* when $D_G(U) \le 6|U|$.

Lemma B.1. *Let G be triangle-free and strong C_4-free, and let $\varnothing \ne U \subseteq V(G)$. Then*

$$D_G(U) = 2e\big(G[U]\big) + 2e(U, V(G) \setminus U),$$

and $\varepsilon_6(G) - \varepsilon_6(G - U) = D_G(U) - 6|U|$. If $D_G(U) \le 6|U|$ and $\varepsilon_6(G) > 0$, then $U \ne V(G)$ and $\varepsilon_6(G - U) \ge \varepsilon_6(G) > 0$. If $K \ge 6$ is real and $D_G(U) \le 6|U|$, then $D_G(U) \le K|U|$.

Proof. By Lemma 2.1, $\lambda(v) = d(v)$ and $S(G) = 2e(G)$. The edges counted by $S(G)$ but not by $S(G - U)$ are exactly the edges that meet U, so $D_G(U) = 2e(G[U]) + 2e(U, V(G) \setminus U)$. The identity for ε_6 is the definition. If $U = V(G)$, then $D_G(U) = S(G)$, hence $D_G(U) - 6|U| = \varepsilon_6(G)$. Thus $\varepsilon_6(G) > 0$ forbids $D_G(V(G)) \le 6|V(G)|$. The inequality $D_G(U) \le 6|U|$ rearranges to $\varepsilon_6(G - U) \ge \varepsilon_6(G)$. For real $K \ge 6$ one has $6|U| \le K|U|$. □

Lemma B.2. *Assume that every nonempty triangle-free strong C_4-free graph of minimum degree at least 4 contains an induced P_{11}. Then every triangle-free strong C_4-free graph with $p \le 10$ is 3-degenerate, and therefore satisfies $e \le 3(n-1)$ and $S \le 6(n-1)$ whenever $n \ge 1$. In particular $S \le 6n$.*

Proof. Let H be a nonempty induced subgraph of a triangle-free strong C_4-free graph with $p \le 10$. Then H is triangle-free and strong C_4-free, and $p(H) \le 10$, so H has no induced P_{11}. The assumed hypothesis forces $\delta(H) \le 3$. Delete a vertex of degree at most 3 in the current graph and repeat. Each of the first $n-1$ deletions removes at most three edges, and the last vertex is isolated, so $e \le 3(n-1)$. Lemma 2.1(3) gives $S = 2e$. □

The path hypothesis of Lemma B.2 is not proved in this paper.

Proposition B.3. *Let $\mathcal{R} \subseteq \mathcal{F}$. Suppose that every positive graph in $\mathcal{F} \setminus \mathcal{R}$ has a nonempty 6-cheap set. Then the following are equivalent.*

1. *Every positive graph in $\mathcal{R}$ has a nonempty 6-cheap set.*
2. *Every graph in $\mathcal{F}$ satisfies $S \le 6n$.*

The supposition is not a proof that any particular subclass is empty.

Proof. If (2) holds, then no graph in $\mathcal{F}$ is positive, so (1) holds vacuously. Conversely, assume (1) and the supposition, and suppose some graph in $\mathcal{F}$ is positive. Among all positive induced subgraphs of members of $\mathcal{F}$, choose H of minimum order. Then $H \in \mathcal{F}$, and H is connected. If H had a nonempty 6-cheap set U, Lemma B.1 would give a smaller positive induced subgraph $H - U$. Thus $H \in \mathcal{R}$, contradicting (1). Hence no graph in $\mathcal{F}$ is positive. □

Let $\mathcal{F}_0$ be the class defined in Section 10. Proposition B.3 does not show that $\mathcal{F}_0$ contains no positive graph.

Appendix C. Ports of a distance-preserving dominating set

Lemma C.1. *Let G be triangle-free and strong C_4-free, and let $D \subseteq V(G)$ be a connected dominating set such that $d_G(x,y) = d_{G[D]}(x,y)$ for all $x, y \in D$. Write $H = V(G) \setminus D$.*

1. *Every vertex of H has exactly one neighbour in D.*
2. *If $x, y \in H$ are adjacent, with neighbours a and b in D, then $a \ne b$, $a \not\sim b$, and $d_{G[D]}(a,b) \le 3$.*

Proof. Let $x \in H$. Since D dominates and $x \notin D$, the vertex x has a neighbour in D. If $a, b \in N(x) \cap D$ are distinct, then $d_G(a,b) \le 2$, hence $d_{G[D]}(a,b) \le 2$. Distance one gives the triangle axb. Distance two gives a 4-cycle through x and a length-2 path in $G[D]$.

For (2), if $a = b$ then axy is a triangle, and if $a \sim b$ then $axyb$ is a 4-cycle. The path a, x, y, b has length three, so $d_{G[D]}(a,b) \le 3$. □

Lemma C.2. *Under the hypotheses of Lemma C.1, the cut between D and H has exactly $|H|$ edges, and*

$$\varepsilon_6(G) = 2e(G[D]) - 6|D| + 2e(G[H]) - 4|H|.$$

If in addition $e(G[H]) \le 2|H|$, then $\varepsilon_6(G) \le 2e(G[D]) - 6|D|$. The inequality $\varepsilon_6(G) \le -2|D|$ does not follow from this identity alone. In particular the identity does not produce a 6-cheap set in every triangle-free strong C_4-free graph of path order 11 or 12.

Proof. Each vertex of H has exactly one neighbour in D. Lemma 2.1(3) gives $S(G) = 2\big(e(G[D]) + e(G[H]) + |H|\big)$. Subtract $6(|D| + |H|)$. □

If D is a single vertex and the minimum degree of G is at least 4, Lemma C.1 forces G to be a star, whose leaves have degree one. Thus no one-vertex set satisfies the hypotheses of Lemma C.1 in a graph of minimum degree at least 4.

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